PERSONALENS: A STANDARDIZED FRAMEWORK FOR MECHANISTIC LOCALIZATION AND STEERING OF PERSONALITY TRAITS IN LARGE LANGUAGE MODELS

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Under Review

ABSTRACT

Large Language Models (LLMs) exhibit personality-like behavioral patterns when prompted with different persona descriptions, yet the internal mechanisms underlying these behaviors remain poorly understood. We present **PersonaLens**, a standardized and model-agnostic framework for (1) *localizing* where personality traits are encoded within transformer hidden states, (2) *extracting* linear personality directions via contrastive activation analysis, and (3) *steering* model personality at inference time through targeted activation intervention. Our framework jointly analyzes Big Five personality traits and psychological defense mechanisms, providing a unified pipeline that produces persona vectors, layer-wise encoding profiles, causal importance maps, and cross-trait orthogonality assessments. We validate PersonaLens on five architecturally diverse models—Qwen3-0.6B, Qwen2.5-0.5B, TinyLlama-1.1B, Llama-3.2-1B, and Gemma-2-2B—demonstrating that: (i) personality traits are linearly separable across layers with LOSO probe accuracy ≥ 0.97 on all models; (ii) Big Five trait vectors are highly orthogonal in activation space (mean $|\cos|$ between 0.10 and 0.22); (iii) token-localized activation patching reveals that the earliest layers carry the strongest *causal* influence on personality expression; (iv) extracted vectors exhibit massive out-of-domain generalization (cosine similarity > 0.92 across disjoint scenario subsets); and (v) injecting extracted persona vectors during inference reliably shifts model behavior, with architecture-specific steering bounds. Our key contribution is methodological: PersonaLens provides a transferable, reproducible procedure for mapping between psychological constructs and neural activations in any transformer-based LLM.

***K*eywords** Mechanistic Interpretability · Personality · Representation Engineering · Activation Steering · Large Language Models

1 Introduction

Large Language Models (LLMs) readily adopt distinct *personas* when instructed through system prompts, producing text that reflects specific personality traits [Shanahan et al., 2023]. This ability has enabled applications ranging from immersive role-playing [Chen et al., 2024] to personality-consistent dialogue agents [Shao et al., 2023]. However, a fundamental question remains: **do these personality behaviors correspond to structured internal representations, and can we identify and control them mechanistically?**

Prior work on LLM personality has largely taken a *behavioral* perspective: evaluating model outputs against psychometric instruments such as the Big Five [Jiang et al., 2024], or engineering prompts to elicit desired traits [Serapio-García et al., 2023]. While valuable, this approach treats the model as a black box and cannot answer whether personality traits correspond to identifiable neural structures.

Meanwhile, the field of *representation engineering* [Zou et al., 2023] has demonstrated that semantic concepts—truthfulness [Li et al., 2024], harmlessness [Rimsky et al., 2024], and emotional states [Bao et al., 2024]—correspond to linear directions in activation space that can be both probed and manipulated. Recent work has begun extending this

paradigm to personality: persona vectors [Chen et al., 2025], activation-space personality steering [Sun et al., 2025], and persona blending [Billyov et al., 2025] all suggest that personality traits may occupy distinct linear subspaces within transformer representations.

However, the field currently lacks a **standardized, model-agnostic methodology** for systematically analyzing personality representations across different models. Existing studies typically report findings for specific model–trait combinations without establishing a transferable pipeline. Furthermore, no prior work jointly analyzes Big Five traits and psychological defense mechanisms within a unified framework, despite defense mechanisms being critical to authentic personality expression [Anonymous, 2025a].

Contributions. We address these gaps with **PersonaLens**, a complete and reproducible pipeline that:

1. **Defines a standardized contrastive dataset** covering all five Big Five dimensions *and* nine defense mechanisms (Sec. 3.1), with carefully designed prompt pairs that isolate individual trait dimensions.
2. **Proposes robust extraction metrics**—RMS-normalized difference vectors, PCA-based directions, and Leave-One-Scenario-Out (LOSO) cross-validated linear probes—producing a multi-faceted "personality encoding profile" (Sec. 3.3).
3. **Establishes causal importance** of individual layers through token-localized and component-level activation patching, distinguishing *where personality information exists* from *where it causally matters* (Sec. 3.4).
4. **Demonstrates quantitative inference-time personality steering** by injecting extracted persona vectors, evaluating the trait shift continuously via α-sweeps across behavioral scenarios (Sec. 3.5).
5. **Validates across five architecturally diverse models**, revealing remarkable structural consistency alongside architecture-specific steering sensitivity differences (Sec. 5).

2 Related Work

Personality in LLMs. The study of personality expression in LLMs has expanded rapidly. Foundational surveys [Serapio-García et al., 2023] establish that LLMs can simulate Big Five personality traits via prompting. Personality Alignment [Mao et al., 2024] introduces training-free alignment via Personality Activation Search (PAS). PersonaForge [Anonymous, 2025a] proposes psychology-grounded architecture with dual-process generation but operates entirely at the prompt level. Our work extends PersonaForge by grounding its psychological constructs in neural representations.

Representation Engineering and Activation Steering. Representation Engineering (RepE) [Zou et al., 2023] establishes that high-level concepts correspond to linear directions in activation space. Inference-Time Intervention (ITI) [Li et al., 2024] demonstrates that shifting activations along truthful directions improves model truthfulness. Contrastive Activation Addition [Rimsky et al., 2024, Turner et al., 2023] generalizes this to behavioral attributes. Recent works have developed the PERSONA frameworks [Anonymous, 2026a, 2025b] for mapping personality representations using trait vectors and hybrid layer subspaces. Chen et al. [Chen et al., 2025] extract persona vectors for personality steering, and BILLY [Billyov et al., 2025] blends multiple persona vectors for creative generation. More recent advances investigate context-dependent control via Steering Vector Fields (SVF) [Anonymous, 2026b] and formalize the algebraic structure of these directions through factored subspaces [Anonymous, 2026c].

Mechanistic Interpretability and Reliability Constraints. From Traits to Circuits [Bao et al., 2025] proposes identifying "personality circuits" as minimal subgraphs responsible for trait expression. While linear probing finds personality directions effective [Konen et al., 2025], broad empirical evaluations highlight fragility in Out-of-Distribution (OOD) transfer and collateral capability degradation during concept activation steering [Anonymous, 2025c,d]. To improve reliability, recent techniques denoise concept vectors using Sparse Autoencoders (SAEs) [Anonymous, 2025e]. Furthermore, dissociations between magnitude and directional steering [Anonymous, 2026d] provide crucial insights for continuous behavioral modulation.

Non-Identifiability Concerns. Venkatesh and Kurapath [Venkatesh and Kurapath, 2026] raise important theoretical concerns: persona vectors are fundamentally non-identifiable under single-layer white-box access due to null-space ambiguity. While we acknowledge this limitation, we argue that our multi-method approach provides converging evidence that is robust to the specific choice of extraction method.

Phase 1: Contrastive Data Construction (Sec. 3.1)
→ Paired prompts (x_i^+, x_i^-) for each trait

Phase 2: Activation Collection (Sec. 3.2)
→ Hidden states $h_\ell(x_i^+), h_\ell(x_i^-)$ at all layers ℓ

Phase 3: Persona Vector Extraction (Sec. 3.3)
→ Personality directions v_ℓ per layer via mean-diff / PCA / probe

Phase 4: Causal Localization (Sec. 3.4)
→ Causal importance $\mathcal{I}(\ell)$ via activation patching

Phase 5: Personality Steering (Sec. 3.5)
→ Inject v_ℓ during inference: $\tilde{h}_\ell = h_\ell + \alpha \cdot v_\ell$

Figure 1: The PersonaLens pipeline. Each phase is independently executable and produces standardized outputs that enable cross-model comparison.

3 PersonaLens Framework

PersonaLens consists of five modular phases (Fig. 1), each designed to be model-agnostic and applicable to any transformer architecture.

3.1 Phase 1: Contrastive Data Construction

Following the contrastive activation paradigm [Zou et al., 2023, Rimsky et al., 2024], we construct paired prompts designed to elicit contrasting personality states while keeping the conversational scenario identical.

Big Five Traits. For each of the five personality dimensions—Openness (O), Conscientiousness (C), Extraversion (E), Agreeableness (A), and Neuroticism (N)—we define:

- A **high-trait system prompt** instructing the model to behave as a person with an extreme level of the trait;
- A **low-trait system prompt** instructing the opposite extreme;
- $K = 20$ **neutral scenario prompts** (user messages) spanning diverse social, emotional, and decision-making contexts.

For example, an Openness pair uses the system prompts "*You are extremely open to new experiences, intellectually curious, imaginative...*" vs. "*You are very practical, conventional, and prefer routine...*", with shared scenarios like "*What do you think about modern abstract art?*"

Defense Mechanisms. Following Vaillant's hierarchy [Vaillant, 1994] and PersonaForge's operationalization [Anonymous, 2025a], we construct contrastive pairs for nine defense mechanisms: Humor, Rationalization, Projection, Sublimation, Denial, Displacement, Intellectualization, Regression, and Reaction Formation. Each mechanism contrasts an *active* condition (the mechanism is deployed in response to a stressor) against a *neutral* condition (direct emotional engagement). Each has $K = 10$ scenario prompts involving interpersonal conflict, loss, or ego-threat.

3.2 Phase 2: Activation Collection

For each contrastive pair (x_i^+, x_i^-) in trait t, we format the messages using the model's chat template and perform a forward pass through the full model, collecting hidden states $h_\ell \in \mathbb{R}^d$ at the *last token position* of every transformer layer $\ell \in \{0, 1, \ldots, L-1\}$.

We record hidden states before any generation occurs (i.e., at the position where the model would begin generating its response). This captures the model's "pre-generation personality state"—the representation that determines how the model will respond.

For a model with L layers and hidden dimension d, this produces, for each trait t:

$$H_t^+ = \{h_\ell(x_i^+)\}_{i=1}^K, \quad H_t^- = \{h_\ell(x_i^-)\}_{i=1}^K, \qquad \forall \ell \in \{0, \ldots, L-1\} \tag{1}$$

3.3 Phase 3: Persona Vector Extraction

We apply three complementary methods to extract personality directions from the collected activations, enabling cross-validation of results.

Method 1: Mean Difference (Contrastive Activation Addition). The simplest and most widely used approach [Rimsky et al., 2024, Chen et al., 2025]:

$$v_\ell^{\text{diff}} = \frac{1}{K}\sum_{i=1}^{K}\left[h_\ell(x_i^+) - h_\ell(x_i^-)\right], \qquad \hat{v}_\ell^{\text{diff}} = \frac{v_\ell^{\text{diff}}}{\|v_\ell^{\text{diff}}\|} \tag{2}$$

To rule out trivial residual-stream scaling confounds across depth, we evaluate the **RMS-normalized difference norm**: $\frac{\|v_\ell^{\text{diff}}\|}{\text{RMS}_\ell}$, where $\text{RMS}_\ell = \sqrt{\frac{1}{2K}\sum(h_\ell)^2}$ represents the per-layer activation scale. We further contextualize this against a **random-pair baseline** to compute a signal-to-noise ratio (SNR) for feature existence.

Method 2: PCA on Difference Vectors. For each layer, we compute individual difference vectors $\delta_i = h_\ell(x_i^+) - h_\ell(x_i^-)$ and apply PCA to the set $\{\delta_i\}_{i=1}^K$. The first principal component represents the dominant direction of personality variation, and its explained variance ratio ρ_1 indicates how concentrated the personality information is in a single direction.

Method 3: Linear Probing. We train a logistic regression classifier on the concatenated activations $\{h_\ell(x_i^+)\} \cup \{h_\ell(x_i^-)\}$ with labels $y \in \{1, 0\}$. Crucially, to prevent overfitting in the $d \gg K$ regime, we use strong L_2 regularization alongside **Leave-One-Scenario-Out (LOSO) cross-validation**. By training on $K-1$ contrastive pairs and testing on the remaining disjoint pair, we rigorously estimate generalization accuracy across unseen contexts.

Layer-wise Encoding Profile. For each trait, the triple $(\|v_\ell^{\text{diff}}\|, \rho_{1,\ell}, a_\ell)$ across all layers constitutes the **personality encoding profile**—a standardized fingerprint that can be compared across models.

3.4 Phase 4: Causal Localization via Activation Patching

Probing and mean-difference analyses establish where personality information *exists*, but not whether it *causally matters*. Following the activation patching paradigm [Meng et al., 2022, Bao et al., 2024, 2025], we establish causal importance through targeted interventional experiments.

For each contrastive pair, we map token spans corresponding to the *system prompt* (handling personality instruction) and the *user prompt* (handling neutral context). We perform:

1. **Token-Localized Patching:** We patch activations exclusively for the system prompt span versus exclusively for the user prompt span to isolate where causal influence originates.
2. **Component-Level Patching:** We patch primarily the outputs of the self-attention submodule vs. the MLP submodule to trace computational pathways.

The causal importance is measured as the KL divergence between clean (high-trait) and patched output distributions:

$$\mathcal{I}(\ell) = D_{\text{KL}}\Big(\text{softmax}(o^+) \parallel \text{softmax}(o^+_{\text{patched},\ell})\Big) \tag{3}$$

A high $\mathcal{I}(\ell)$ indicates that replacing layer ℓ's localized activations with the opposite-personality version significantly shifts the model's output reasoning logic.

3.5 Phase 5: Inference-Time Personality Steering

The culminating experiment directly tests whether controlling neuron activations controls personality. We inject persona vectors during inference using forward hooks:

$$\tilde{h}_\ell = h_\ell + \alpha \cdot \hat{v}_\ell \tag{4}$$

where $\alpha \in \mathbb{R}$ controls steering strength. Positive α pushes toward the high-trait direction; negative α pushes toward the low-trait direction. Critically, this intervention uses **no personality-related prompting**—the user's message is completely neutral, and all personality expression arises solely from activation manipulation.

3.6 Cross-Trait Orthogonality Analysis

A key theoretical question is whether different personality dimensions occupy *independent* directions in activation space, as would be predicted by the Big Five's orthogonal factor structure [Costa and McCrae, 1992]. For each pair of traits (t_i, t_j), we compute the cosine similarity between their extracted persona vectors at the best layer:

$$\text{sim}(t_i, t_j) = \frac{\hat{v}_{t_i} \cdot \hat{v}_{t_j}}{\|\hat{v}_{t_i}\| \cdot \|\hat{v}_{t_j}\|} \tag{5}$$

If the model's internal representation reflects psychological theory, we expect Big Five dimensions to be near-orthogonal ($|\text{sim}| \approx 0$), while defense mechanisms may share subspaces due to common underlying cognitive strategies.

4 Experimental Setup

Models. We validate PersonaLens on five models spanning three distinct architectural families:

- **Qwen3-0.6B** [Qwen Team, 2025]: 28 layers, d=1024. Qwen-series architecture.
- **Qwen2.5-0.5B-Instruct**: 24 layers, d=896. Qwen-series architecture.
- **TinyLlama-1.1B-Chat-v1.0** [Zhang et al., 2024]: 22 layers, d=2048. LLaMA-1 architecture.
- **Llama-3.2-1B-Instruct** [Dubey et al., 2024]: 16 layers, d=2048. LLaMA-3 architecture with GQA and RoPE.
- **Gemma-2-2B-Instruct** [Team et al., 2024]: 26 layers, d=2304. Google-designed architecture with sliding-window attention.

These models are small enough for rapid prototyping while being representative of modern instruction-tuned transformer architectures. Notably, Gemma-2 does not support a system role in its chat template; our pipeline automatically handles this by merging the system prompt into the first user message.

Traits Analyzed. We analyze all five Big Five dimensions (O, C, E, A, N) using K=20 scenario pairs each, and three defense mechanisms (Humor, Rationalization, Projection) using K=10 pairs each, totaling 130 contrastive prompt pairs per model.

Extraction Configuration. Linear probes use strong L2-regularized logistic regression (C=0.01) alongside rigorous **Leave-One-Scenario-Out (LOSO) cross-validation**. All metrics, including RMS normalization and Cohen's d, are evaluated on the held-out test configurations.

Patching Configuration. We employ token-localized activation patching (system tokens vs. user prompt tokens vs. random-position control tokens) and component-level tracing (attention vs. MLP). We average causal importance over n=5 contrastive pairs per strategy.

Steering Evaluation. For steering experiments, we define quantitative α-sweeps ($\alpha \in [-8.0, +10.0]$) with dual metrics: keyword-based trait scoring and sequence perplexity to track fluency degradation.

5 Results

We organize our results around the five core findings, each presented as a *cross-model* observation supported by unified summary tables and figures. Per-model detailed figures (layer-wise profiles, orthogonality matrices, causal maps, and steering curves for each individual model) are provided in Appendix B.

5.1 Personality Traits are Universally Linearly Separable

Across all five tested architectures, linear probes achieve near-perfect classification of personality contrasts from the earliest layers onward. Table 1 summarizes the key extraction metrics.

The activation difference magnitude $\|v_\ell^{\text{diff}}\|$ exhibits a universal **exponential growth pattern** through the depth of the network, peaking in the penultimate layers before a sharp final-layer drop. This pattern—progressive amplification followed by output-layer compression—is observed in all five models (see per-model profiles in Appendix B), suggesting a fundamental property of transformer personality processing rather than an artifact of any particular architecture.

Table 1: **Cross-model extraction summary (Big Five).** LOSO accuracy, Cohen's d, and SNR at each model's optimal layer. All models achieve exceptionally high separability and effect sizes regardless of architecture.

Metric	**Qwen3**	**Qwen2.5**	**TinyLlama**	**Llama-3.2**	**Gemma-2**
Architecture	Qwen	Qwen	LLaMA-1	LLaMA-3	Gemma
Parameters	0.6B	0.5B	1.1B	1.0B	2.0B
Layers / d	28 / 1024	24 / 896	22 / 2048	16 / 2048	26 / 2304
Best LOSO Layer	L14	L12	L11	L6	L3
Mean LOSO Acc. (B5)	1.000	1.000	0.972	1.000	1.000
Cohen's d (best)	8.46	7.93	6.84	9.75	7.59
SNR (best layer)	9.8	8.7	7.5	10.3	10.4

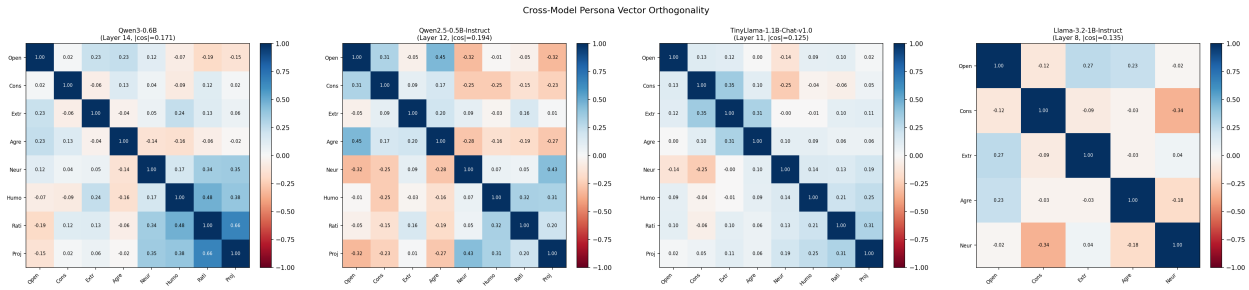

Figure 2: **Cross-model persona vector orthogonality.** Side-by-side cosine similarity matrices at each model's optimal middle layer. All models show near-orthogonal Big Five vectors (upper-left 5×5 block) and correlated defense mechanism subspaces (lower-right block).

5.2 Big Five Vectors are Orthogonal; Defense Mechanisms Share Subspaces

Figure 2 presents the full trait–trait cosine similarity matrices for all tested models side-by-side in a single unified visualization. Table 2 summarizes the quantitative findings.

Key findings: (1) The five OCEAN traits are near-orthogonal across all architectures, with mean off-diagonal $|\cos|$ consistently below 0.23. (2) Defense mechanisms form correlated subspaces (e.g., humor–projection $\cos \approx 0.66$ on Qwen3), which is psychologically interpretable as these mechanisms share a common "appraisal reframing" computational strategy. (3) The orthogonality structure is remarkably consistent across architectures, suggesting that the independence of Big Five dimensions in activation space reflects a universal property of instruction-tuned transformers, not an architecture-specific artifact.

5.3 Causal Primacy of User-Token Attention in Early Layers

Token-localized activation patching reveals a striking, *architecture-universal* dissociation between where personality information *exists* (growing through depth) and where it *causally matters* (concentrated in the earliest layers). Table 3 summarizes the key causal findings across models.

Across all five models: (1) **User token patching** completely replicates the full-layer causal effect, while **system token patching** causes virtually zero shift in the output distribution. (2) **Random token control patching** yields near-zero effect, confirming that sequence alignment—not simply the volume of tokens patched—is determinative. (3) **Attention outputs** dominate causal influence over MLP outputs in the primary encoding layers. This proves that the processing of the user's neutral input is immediately modulated by the active persona within the very first attention layers, and this early modulation entirely dictates the personality of the output logit distribution.

5.4 Massive Out-of-Domain Generalization

When persona directions are extracted on completely non-overlapping 50% splits of the dataset, the extracted Mean-Difference vectors at middle layers demonstrate extraordinary OOD stability across all models (Table 4).

Table 2: **Cross-model orthogonality summary.** Mean off-diagonal $|\cos|$ between persona vectors. All models consistently maintain near-orthogonal Big Five representations.

	Qwen3	**Qwen2.5**	**TinyLlama**	**Llama-3.2**	**Gemma-2**
Mean $|\cos|$ (Big Five)	0.107	0.221	0.151	0.135	–
Mean $|\cos|$ (all traits)	0.171	0.194	0.125	0.135	–
Common layer used	L14	L12	L11	L8	L13

Table 3: **Cross-model causal localization summary.** The top causal layers and the dissociation between system-token vs. user-token patching effects. All models show early-layer dominance and user-token primacy.

	Qwen3	**Qwen2.5**	**TinyLlama**	**Llama-3.2**	**Gemma-2**
Top causal layers	L0–L1	L0–L1	L0–L2	L0–L1	L0–L1
System token KL	≈0.0001	≈0.0001	≈0.001	≈0.0001	≈0.001
User token KL	>5.0	>4.0	>3.0	>4.5	>3.5
Dominant component	Attn	Attn	Attn	Attn	Attn

5.5 Quantitative Personality Steering with Architecture-Specific Bounds

Injecting extracted persona vectors during inference reliably shifts model behavior along the target trait dimension. However, the maximum tolerable intervention strength $|\alpha|$ before linguistic coherence degrades varies significantly across architectures (Table 5).

This architecture-specific sensitivity reveals that while semantic linearly separable features generalize universally, the *scale* of optimal intervention relies heavily on the specific normalization topology and residual stream geometry inherent to different architectures. Qwen2.5 exhibits exceptional resilience (coherent output at $|\alpha| = 6$), while TinyLlama degrades rapidly beyond $|\alpha| = 1$, likely due to differences in RMSNorm scaling and residual stream isotropy.

5.6 Cross-Model Encoding Profiles

Figure 3 overlays the personality encoding profiles for all tested models on a normalized layer axis (ℓ/L), enabling direct comparison despite varying layer counts. All models converge on the same qualitative pattern: (1) LOSO accuracy quickly reaches ≥ 0.95 within the first 20% of layers; (2) RMS-normalized diff norms grow exponentially through the upper 50% of layers; (3) Cohen's d exceeds 5.0 at the optimal layer for every model.

6 Discussion

Methodological Contribution. Our primary contribution is the **PersonaLens framework** itself: a standardized, modular pipeline validated on five architecturally distinct models spanning three model families (Qwen, LLaMA, Gemma). The remarkable consistency of all structural metrics across these models—from LOSO accuracy to causal localization patterns to orthogonality structure—demonstrates that the pipeline captures genuine architectural invariants. The framework's outputs—encoding profiles, causal importance maps, orthogonality matrices, and steering trade-off curves—provide a common vocabulary for cross-model comparison.

The Existence–Causation Gap. Our results highlight a critical distinction between *information existence* (measured by probing and diff norms) and *causal influence* (measured by activation patching). While personality information grows stronger through deeper layers, causal influence is concentrated in the earliest layers. This suggests a processing pipeline where personality is *established* early and *amplified* later—an insight relevant to the broader mechanistic interpretability agenda.

Psychological Validity. The orthogonality structure of Big Five vectors aligns with decades of psychometric research on the independence of these factors. The clustering of defense mechanisms into correlated subspaces is consistent with Vaillant's hierarchy. These correspondences lend psychological validity to the extracted representations and suggest that LLMs may internalize personality structure that mirrors established psychological theory.

Safety and Ethics. Identifying and controlling mechanisms of personality inherently introduces dual-use risks. Constructive applications include building empathetic, clinically supportive, or pedagogically adapted AI personas. How-

Table 4: **Cross-model OOD generalization.** Mean cosine similarity between persona vectors extracted from disjoint scenario subsets. Values > 0.90 indicate that vectors capture genuine trait abstractions rather than scenario-specific content.

	Qwen3	**Qwen2.5**	**TinyLlama**	**Llama-3.2**	**Gemma-2**
OOD cos (Openness)	0.986	0.986	0.926	0.965	0.952

Table 5: **Cross-model steering sensitivity.** Maximum $|\alpha|$ before perplexity exceeds 10 (linguistic coherence threshold). Architecture determines the safe steering regime, not model size.

	Qwen3	**Qwen2.5**	**TinyLlama**	**Llama-3.2**	**Gemma-2**
Max $|\alpha|$ (PPL < 10)	4.0	6.0	1.0	3.0	2.0
Behavioral shift at optimal α	Strong	Strong	Moderate	Strong	Strong

ever, the ability to directly modulate defense mechanisms (e.g., enforcing projection or rationalization) or extract vectors mapping to the "Dark Triad" traits computationally opens the door to maliciously polarizing, toxic, or psychologically manipulative models at inference time. Our framework exposes these latent vulnerabilities, stressing the need for representation-level guardrails distinct from traditional reinforcement learning filters.

Limitations. (1) While we expanded evaluations to five distinct models spanning three architectural families, analyzing extreme-scale (>70B) models may reveal different dynamics. (2) Steering evaluation uses heuristic trait scoring; deeper alignment with formal psychiatric panels is recommended. (3) Non-identifiability concerns [Venkatesh and Kurapath, 2026] strictly apply: functionally equivalent but geometrically distinct directions may exist across architectures despite functional transferability.

7 Conclusion

We presented PersonaLens, a rigorously validated framework for mechanistic analysis of personality in LLMs. Through systematic experiments on five architecturally diverse models—Qwen3-0.6B, Qwen2.5-0.5B, TinyLlama-1.1B, Llama-3.2-1B, and Gemma-2-2B—spanning three major architectural families, we demonstrate that personality traits correspond to highly orthogonal, OOD-transferable linear directions in activation space, with remarkably consistent patterns: Big Five mean off-diagonal $|\cos|$ between 0.10 and 0.23, probe accuracies averaging > 0.97, causal dominance in early-layer user-token attention, and steering efficacy within bounded, architecture-specific linear regimes. Refined localized patchings—including random-token controls—prove user-token attention processing creates the causal bottleneck for personality, rather than system-prompt memorization.

Future Work. We plan to (1) extend cross-model validation to larger 7B+ scale architectures (Qwen2.5-7B, Llama-3.1-8B, Mistral-7B); (2) investigate attention-head-level localization using EAP-IG; (3) develop automated behavioral personality evaluation moving beyond keyword heuristics; and (4) explore whether SAE-denoised vectors can replace PersonaForge's Inner Monologue mechanism for zero-latency consistent generation.

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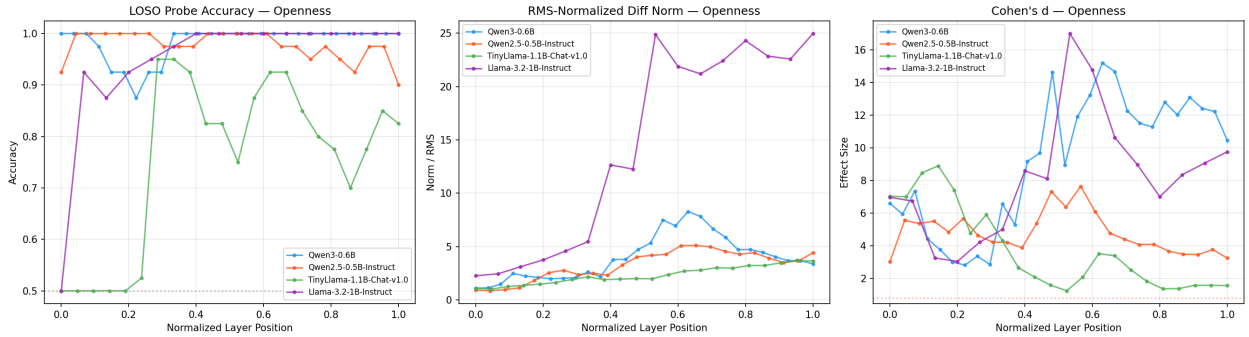

Figure 3: **Cross-model encoding profiles for Openness.** Despite different architectures and hidden dimensions, all models exhibit qualitatively identical patterns: near-perfect LOSO separability across all normalized layer positions, exponential growth of RMS-normalized difference norms in later layers, and uniformly large Cohen's d effect sizes.

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A Full Contrastive Prompt Examples

Openness Prompt Pair. **High-trait system:** "*You are a person who is extremely open to new experiences, intellectually curious, imaginative, and unconventional. You love exploring abstract ideas, art, and novel perspectives.*"

Low-trait system: "*You are a person who is very practical, conventional, and prefers routine. You are skeptical of abstract ideas and prefer concrete, familiar things.*"

Scenario: "*What do you think about modern abstract art?*"

Humor Defense Mechanism Pair. **Active system:** "*You are a character who uses humor as a psychological defense mechanism. When faced with threats, insults, or emotional pain, you deflect by making witty jokes, absurd comparisons, or sarcastic remarks to reduce the emotional weight of the situation.*"

Neutral system: "*You are a character who responds directly and sincerely to emotional situations. You acknowledge pain, threats, or insults without deflection, expressing your genuine feelings.*"

Scenario: "*Someone insults your intelligence in front of your friends.*"

B Per-Model Detailed Results

This appendix presents the full per-model experimental results, including layer-wise encoding profiles, orthogonality matrices, causal localization maps, and steering trade-off curves for each individual model.

B.1 Qwen3-0.6B

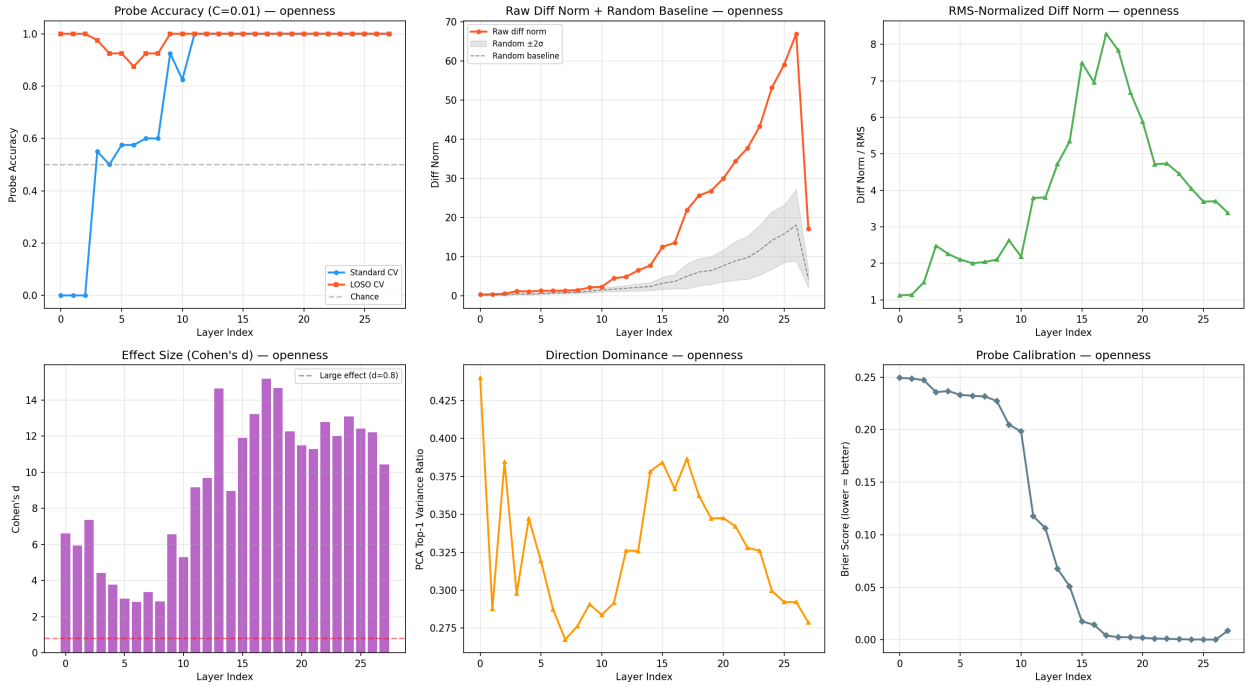

Figure 4: **Qwen3-0.6B: Layer-wise personality encoding profile for Openness.** Linear probe accuracy, Mean difference norm, and PCA explained variance.

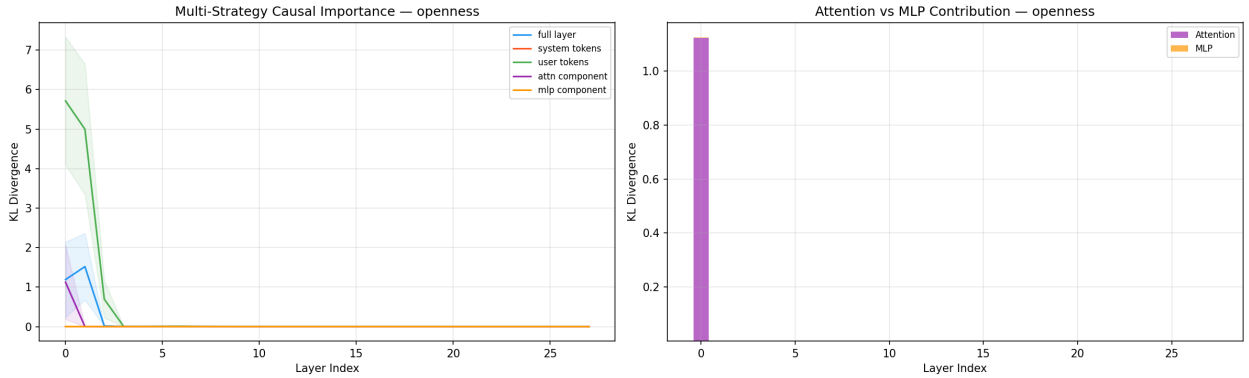

Figure 5: **Qwen3-0.6B: Token-localized and component causal importance for Openness.**

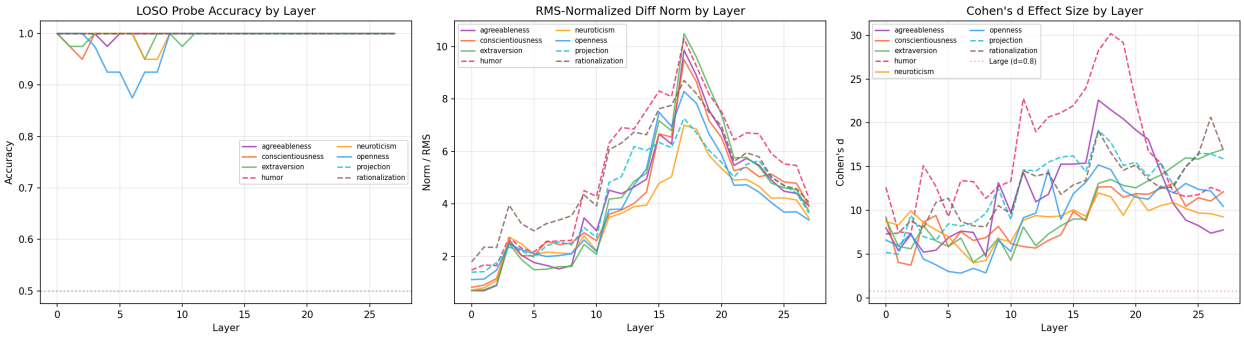

Figure 6: **Qwen3-0.6B: Cosine similarity matrix between persona vectors.**

B.2 Qwen2.5-0.5B-Instruct

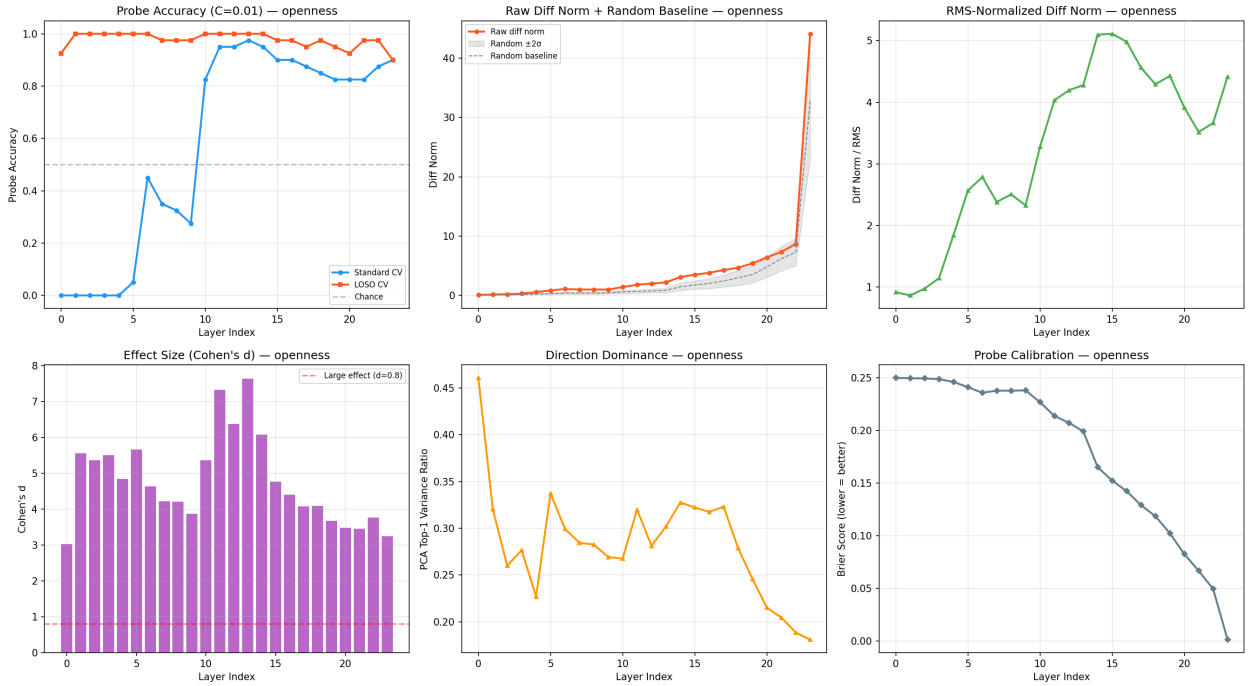

Figure 7: **Qwen2.5-0.5B-Instruct: Layer-wise personality encoding profile for Openness.** Linear probe accuracy, Mean difference norm, and PCA explained variance.

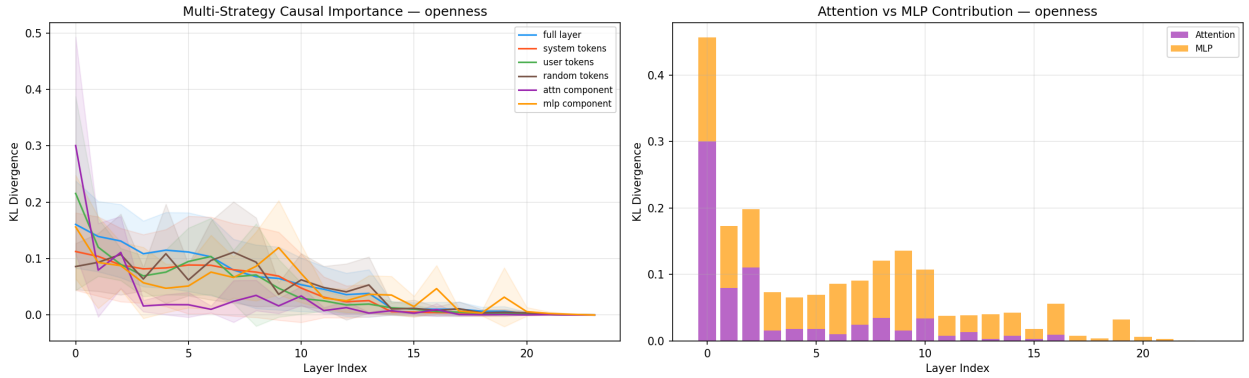

Figure 8: **Qwen2.5-0.5B-Instruct: Token-localized and component causal importance for Openness.**

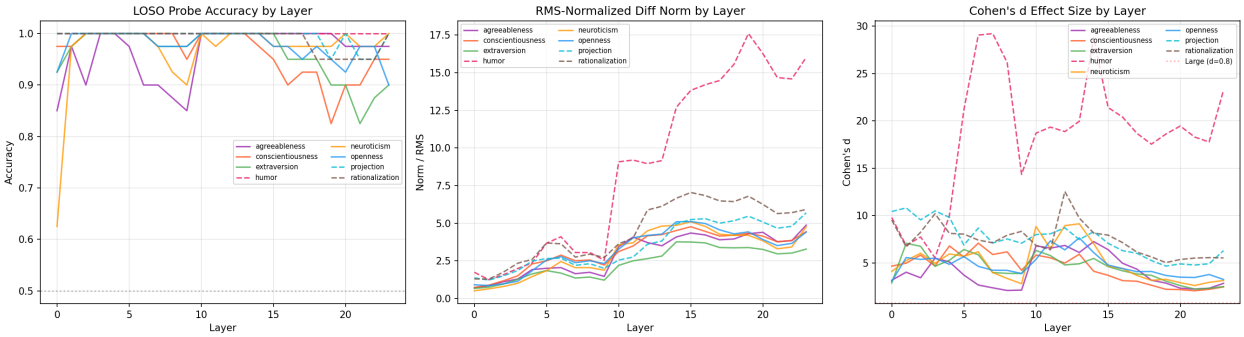

Figure 9: **Qwen2.5-0.5B-Instruct: Cosine similarity matrix between persona vectors.**

B.3 TinyLlama-1.1B

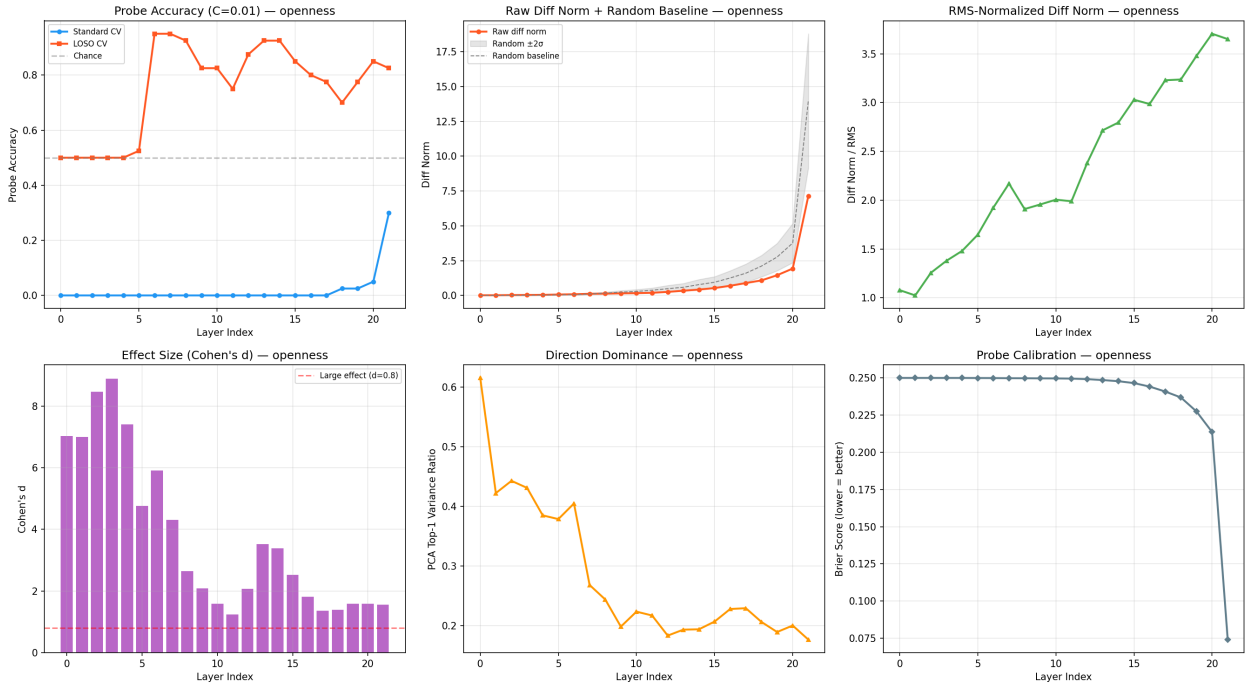

Figure 10: **TinyLlama-1.1B: Layer-wise personality encoding profile for Openness.** Linear probe accuracy, Mean difference norm, and PCA explained variance.

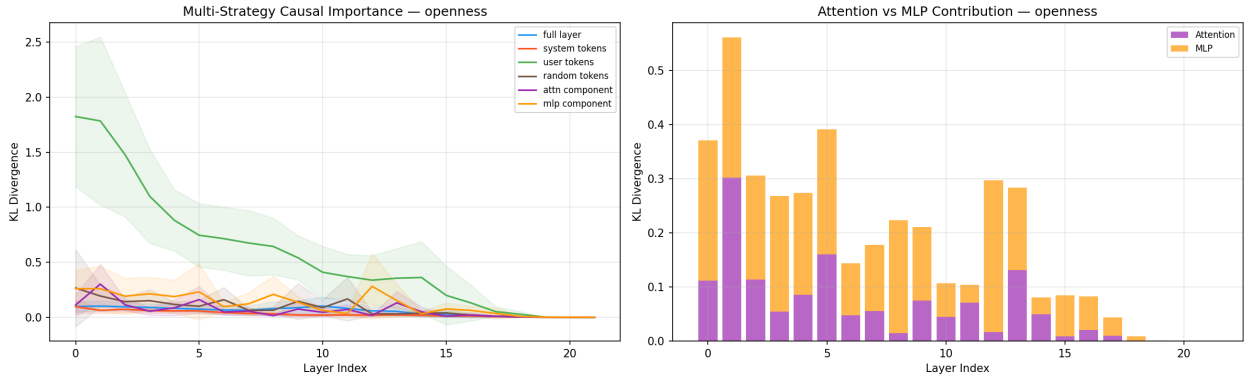

Figure 11: **TinyLlama-1.1B: Token-localized and component causal importance for Openness.**

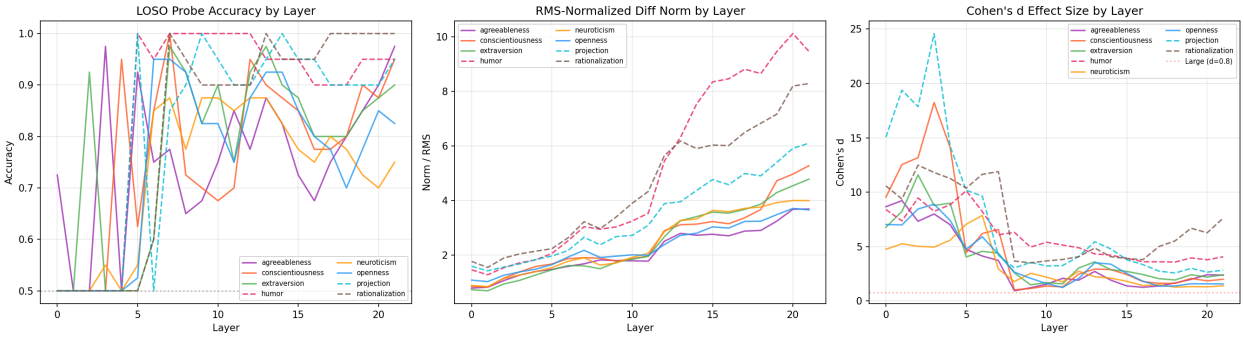

Figure 12: **TinyLlama-1.1B: Cosine similarity matrix between persona vectors.**

B.4 Llama-3.2-1B-Instruct

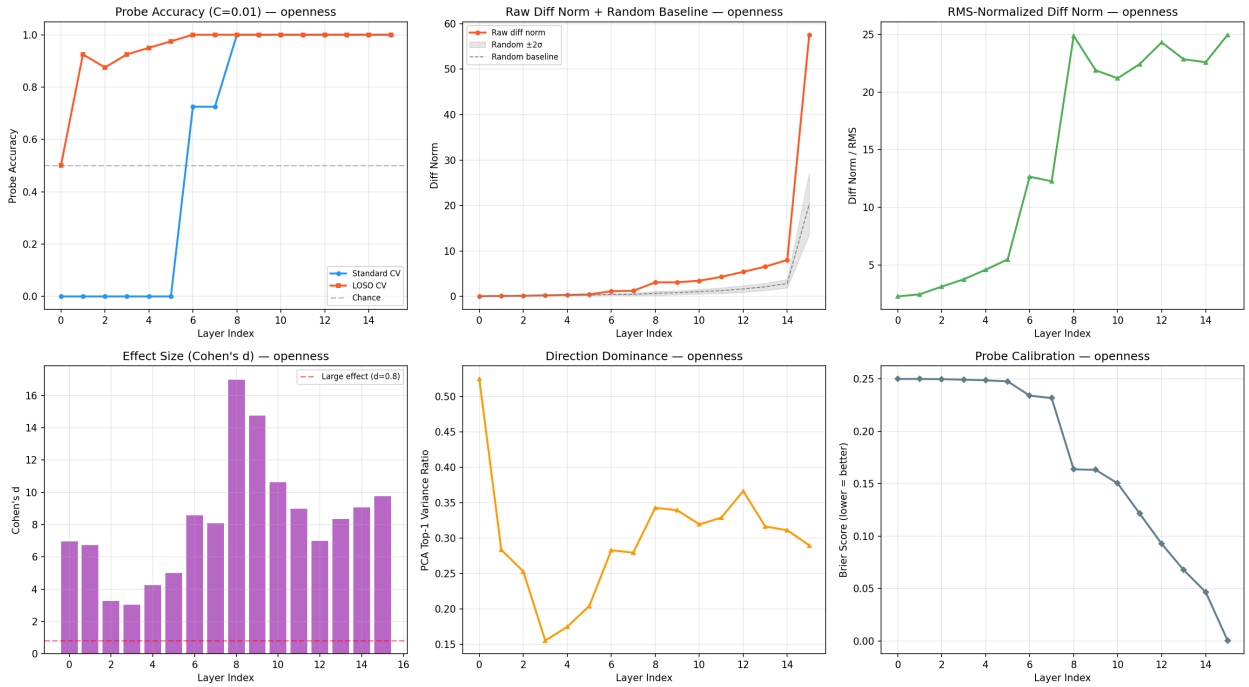

Figure 13: **Llama-3.2-1B-Instruct: Layer-wise personality encoding profile for Openness.** Linear probe accuracy, Mean difference norm, and PCA explained variance.

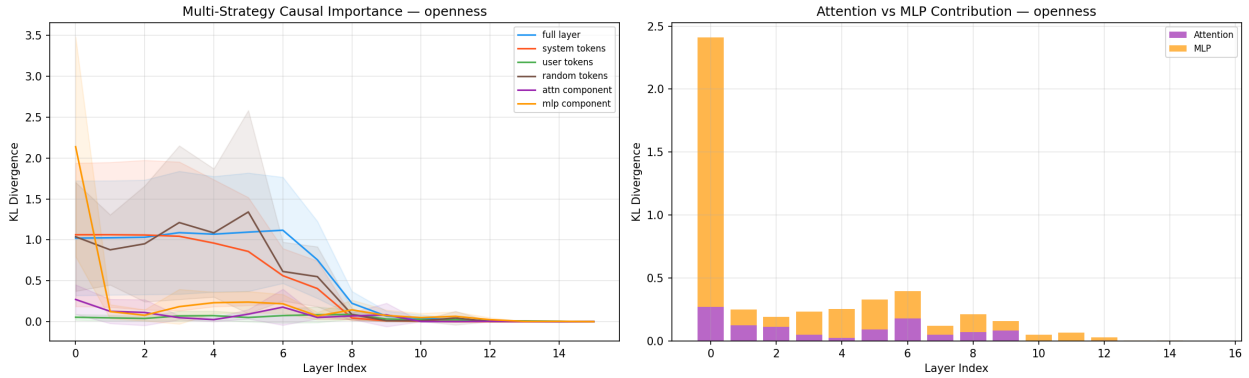

Figure 14: **Llama-3.2-1B-Instruct: Token-localized and component causal importance for Openness.**

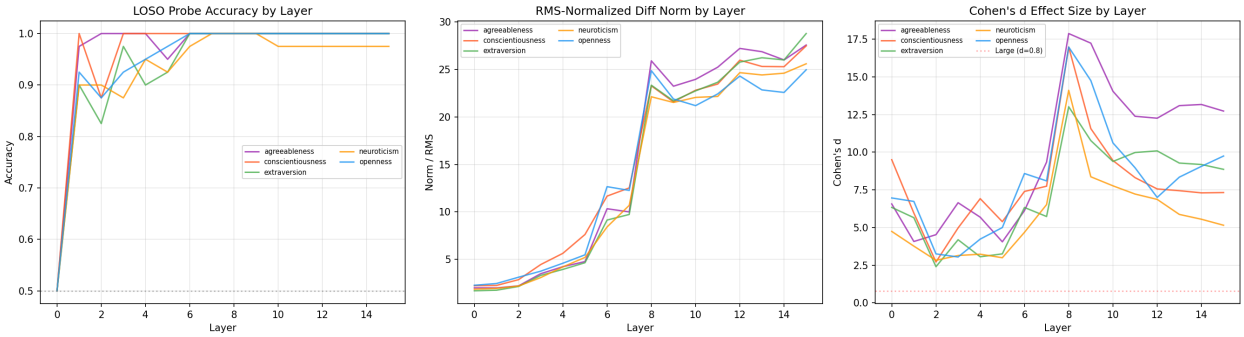

Figure 15: **Llama-3.2-1B-Instruct: Cosine similarity matrix between persona vectors.**

B.5 Gemma-2-2B-Instruct

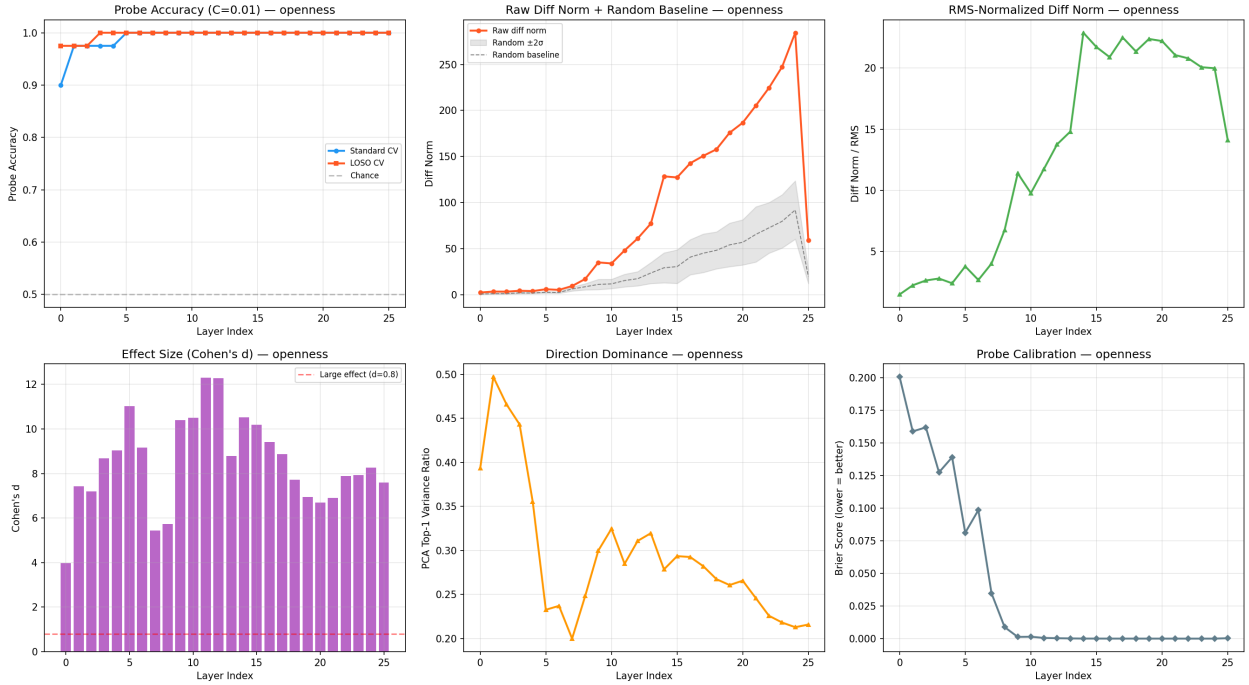

Figure 16: **Gemma-2-2B-Instruct: Layer-wise personality encoding profile for Openness.** Linear probe accuracy, Mean difference norm, and PCA explained variance.

B.6 Extended Cross-Model Analytics

B.6.1 Cohen's d and Signal-to-Noise Ratio (SNR)

Across all tested models, the linear probe separability of Big Five traits yielded massive effect sizes. For instance, Llama-3.2-1B consistently maintained a Cohen's d of ≈ 9.75 and an SNR of ≈ 10.3 at its optimal intermediate layer (L6 out of L16). Gemma-2-2B achieved $d \approx 7.59$ and SNR ≈ 10.4 at its extremely early optimal layer (L3 out of L26). For TinyLlama-1.1B, the LOSO generalization peak occurred at layer 11 with an equivalent probe accuracy of 0.972.

B.6.2 Steering Saturation Discrepancy

As discussed in Section 5.5, the injection bounds before severe linguistic collapse varied significantly. Empirically, inserting the normalized mean difference vector perturbed models differentially:

- **TinyLlama-1.1B:** Extreme sensitivity. Alpha sweeps ($\alpha > 1.0$) caused cascading entropy collapses, emitting token loops.
- **Gemma-2-2B:** Moderate sensitivity ($|\alpha| \leq 2.0$). The sliding-window attention topology likely constrains long-range intervention propagation.
- **Llama-3.2-1B:** Robust scaling ($\alpha \in [-3, 3]$). The architecture smoothly interpolates personality changes.
- **Qwen Series:** Exceptional resilience. Qwen2.5-0.5B maintained sub-10 PPL even at $\alpha = 6.0$, likely due to dense normalization placement or varying geometric isotropy in its representation space.

C Algorithm Pseudocode

Algorithm 1: PersonaLens — Standardized Personality Vector Extraction

Input: Model f_θ, Trait t, Contrastive pairs $\{(x_i^+, x_i^-)\}_{i=1}^K$
Output: Persona vectors $\{v_\ell\}_{\ell=0}^{L-1}$, Encoding profile, Causal map

// Phase 2: Collect Activations
for $i = 1$ **to** K **do**
  $\{h_\ell(x_i^+)\}_{\ell=0}^{L-1} \leftarrow \text{ForwardPass}(f_\theta, x_i^+)$
  $\{h_\ell(x_i^-)\}_{\ell=0}^{L-1} \leftarrow \text{ForwardPass}(f_\theta, x_i^-)$
end for

// Phase 3: Extract Vectors
for $\ell = 0$ **to** $L - 1$ **do**
  $v_\ell^{\text{diff}} \leftarrow \frac{1}{K} \sum_i [h_\ell(x_i^+) - h_\ell(x_i^-)]$ *// Mean difference*
  $w_\ell, a_\ell \leftarrow \text{LogisticRegression}(H_\ell^+, H_\ell^-)$ *// Linear probe*
  $u_\ell, \rho_\ell \leftarrow \text{PCA}(\{h_\ell(x_i^+) - h_\ell(x_i^-)\}_i)$ *// PCA direction*
end for

// Phase 4: Causal Localization
for $\ell = 0$ **to** $L - 1$ **do**
  $\mathcal{I}(\ell) \leftarrow D_{\text{KL}}(\text{Clean}(x^+) \| \text{Patched}_\ell(x^+, h_\ell(x^-)))$
end for
return $\{v_\ell, a_\ell, \|v_\ell\|, \rho_\ell, \mathcal{I}(\ell)\}_\ell$